

**What multiplexing schemes are used in GSM for what purposes? Think also of other layers apart from the physical layer.**

FDD (uplink vs downlink): Separate UL/DL bands (e.g., 890–915 vs 935–960 MHz).

FDMA (across spectrum): Split the band into 200 kHz carriers (124 per direction in GSM-900).

TDMA (within a carrier): 8 time slots per 200 kHz carrier → a “physical channel” is one slot on one carrier.

Burst structure (in a slot): user data is wrapped with training sequence and guard spaces → supports equalization & timing.

PCM/SS7 multiplexing in the core: Traffic/signaling mapped to 64 kbit/s timeslots over 2.048 Mbit/s PCM; interfaces Abis (16 kbit/s) and A (64 kbit/s) are standardized/open.

Packet domain (GPRS): radio MAC/RLC statistically multiplex users over PDCHs; in the core, user IP is tunneled with GTP between SGSN↔GGSN.

**How is synchronisation achieved in GSM? Who is responsible for synchronisation and why is synchronisation very important?**

- **How:** bursts carry a **training sequence** (channel estimation + fine timing) and a **guard space** to absorb residual misalignment; the network commands mobiles to align their transmit timing so each burst lands in its slot.
- **Who:** the BSS (BTS/BSC) controls radio resources and maps **Um**→**A**, i.e., it's responsible for keeping radio channels time-aligned and usable.
- **Why important:** without tight timing, adjacent users' bursts would overlap (inter-slot interference), breaking decoding. The **guard** exists precisely to prevent overlap.

**How is the shift from voice-only to Internet-style traffic reflected in the GSM system?**

**GSM CSD limits:** data standardized at **9.6 kb/s** (14.4 with advanced coding) → not enough for Internet/multimedia.

**Two paths on slides:**

- **HSCSD:** bundle several time slots → higher **AIUR** (e.g.,  $4 \times 14.4 = 57.6$  kb/s); *ready to use, constant quality* but **blocks channels**.
- **GPRS:** **packet switching**, uses **free slots only when packets are ready**; “one step towards UMTS”, but **needs new hardware**.

**Why is a new infrastructure needed for GPRS, but not for HSCSD? Which components are new and what is their purpose?**

HSCSD stays in the circuit-switched core → mainly software/config updates.

GPRS adds packet core nodes: **SGSN** (mobility, billing, security for packet users), **GGSN** (interworking to **PDN**), plus **GR** (addresses). Hence **more investment (new hardware)**.

**Only few operators offered HSCSD – why?**

Because its **bundled circuit slots** are **blocked continuously** → poor spectral efficiency for bursty Internet traffic and reduced voice capacity. (Slide lists “channels blocked for voice transmission”.)

**What are the reasons for the delays in a GSM system for packet data traffic? Distinguish between circuit-switched and packet-switched transmission.**

**Circuit-switched (CSD/HSCSD):** one-off **call setup** delay; afterward, latency is steady but limited by interleaving/coding—no packet scheduling.

**Packet-switched (GPRS):** extra delay from **RACH/TBF setup**, **RLC/MAC scheduling**, **ARQ retransmissions**, **LLC/SNDCP/GTP headers**, and traversal through **SGSN/GGSN**—plus you only transmit when slots are free.

### Relatively high delay, low reliability – why was GPRS yet a break-through in the GSM world?

It introduced **packet, always-on-style access** using spare capacity (“**free slots only if data packets ready**”), enabling IP services and serving as **a step toward UMTS**—even with modest rates and higher delay.

### What are the limitations of a GSM cell in terms of diameter and capacity (voice, data) for the traditional GSM, HSCSD, GPRS? How can the capacity be increased?

**Diameter:** cells vary from **some 100 m up to ~35 km** (terrain, power, user density).

**Voice capacity:** per 200 kHz **carrier** you have **8 time slots**; capacity scales with number of carriers/TRXs and reuse planning; control channels also consume slots.

**HSCSD capacity impact:** each data user **ties up multiple circuit slots** → fewer voice circuits remain.

**GPRS capacity:** shared PDCHs; user throughput depends on available **free slots** and coding.

**Increase capacity by:** adding carriers/TRXs, tighter frequency reuse, sectorization/antenna patterns, and shifting best-effort traffic to **GPRS** so slots are **statistically multiplexed**.

### What determines the coding scheme?

The radio **quality/conditions** (C/I, error rate) and service needs; GPRS/EGPRS uses **modulation & coding schemes** selected by the network to trade robustness vs throughput.

### Give reasons for a handover in GSM and the problems associated with it. Which are the typical steps for handover, what types of handover can occur? Which resources need to be allocated during handover for data transmission using HSCSD or GPRS respectively? What about QoS guarantees?(16-17(3-A))

**Reasons:** falling level/quality in serving cell, better neighbor, interference/load; use **HO\_MARGIN** to avoid ping-pong.

**Types (4):** intra-cell; inter-cell (intra-BSC); inter-BSC (intra-MSC); inter-MSC.

**Typical steps:** MS measurement report → **HO decision** (BSC/MSC) → **resource allocation** in target → **channel activation/ack** → **HO command** to MS → **HO access** → **HO complete** → **clear** old resources.

**Why it's “hard”:** **break-before-make** in GSM (brief interruption).

### Resources during HO:

- **HSCSD:** reserve a **bundle of time slots** in the target cell (same number/class) before command, or call degrades/drops.
- **GPRS:** (re)assign **PDCHs/TBFs** and maintain the **PDP context** via SGSN; scheduling resumes after access.

**QoS guarantees:** classic GSM voice has circuit QoS once set; **GPRS is best-effort** (slides explicitly mention “relatively high delay, low reliability” in the Qs/tasks), so strict guarantees are limited.

### Why using a separated, specialized system like TETRA? Why not using the “normal” cellular phone network with some “extras”? There are reasons for and against such an approach!

### Why TETRA (pros):

- Built-in mission-critical features: **group/broadcast calls, sub-second PTT call setup, priority with pre-emption, end-to-end and air-interface encryption, authentication, and Direct Mode (DMO)** for ad-hoc, **infrastructure-independent** operation (with optional **repeaters/gateways**).

### Why not only TETRA (cons):

- **Throughput is narrowband** (V+D only a few kbit/s per slot; even with TEDS typically ~100–500 kbit/s), and **specialized devices are costly**—hence pressure to use broadband LTE for video/apps. Slides even ask “Future of TETRA? Data rates too low ... LTE Public Safety Broadband.”

Why “commercial cellular + extras” (pros):

- Massive **coverage, capacity** and an **ecosystem** for broadband apps; LTE/5G add public-safety features (see below). Slides flag LTE’s evolution and “Public Safety Features (e.g. D2D).”

But caveats for using current LTE/UMTS/GSM (cons):

- Off-the-shelf networks historically lacked **guaranteed priority/pre-emption, instant group PTT** and **off-network (DMO) operation**; disaster congestion and **backhaul dependence** hurt availability—hence 3GPP’s dedicated public-safety features. (3GPP notes LTE needed extra features for public safety use.)
- Modern fixes add **QoS / Priority / Pre-emption (QPP)** controls to manage congestion for first responders.

**What else could be problematic using e.g. current LTE/UMTS/GSM systems?**

**Resilience/coverage** when power/backhaul fail; need **off-network** comms (DMO-like). 3GPP added **ProSe/sidelink** and **Isolated E-UTRAN (IOPS)** to address this.

**Group efficiency** for large talk-groups → use **multicast/broadcast** (MBMS/5G MBS) instead of many unicasts.

**Assured QoS** under load → require **priority/pre-emption** policies and network support (QPP).

**Check the current approaches of 3GPP towards a public safety mobile network! What is a general trend?**

What 3GPP standardized (high level):

- **Mission-Critical Services (MCX) over LTE/5G:**
  - **MCPTT** (mission-critical Push-To-Talk) in **Release 13**; **MCData** and **MCVideo** in later releases—forming a complete voice/data/video suite.
- **ProSe / Sidelink (PC5)** for **device-to-device** discovery/communication (including one-to-many)—crucial for off-network operations.
- **IOPS (Isolated Operation for Public Safety)** so a local LTE RAN can keep working without core/backhaul. (Stage-1 TS 22.346; architecture in TS 23.180.)
- **Priority/Pre-emption & QoS** mechanisms for congestion management on LTE.
- **Toward 5G:** mission-critical over **5G Core**, **network slicing** for PS traffic, and **5G MBS** (multicast/broadcast) for efficient group delivery.
- **Ecosystem maturation:** certification programmes for **MCX** products (GCF/TCCA).

**General trend (what to memorize):**

*Move from narrowband, specialised PMR (e.g., TETRA) to **broadband LTE/5G** carrying standardized **MCX** services, with resilience (**ProSe/IOPS**), **priority & pre-emption**, and **multicast/broadcast**, often in hybrid deployments (dedicated spectrum + commercial networks). Slides already point to “**LTE Public Safety Broadband**” and “Public Safety Features (e.g., D2D).”*

**What defines the 3rd generation? What was the initial idea?**

**Definition (ITU IMT-2000):** a family of radio interfaces meeting targets around **144 kb/s (vehicular)**, **384 kb/s (pedestrian/suburban)**, up to **~2 Mb/s (indoor/urban)**, plus global roaming and multimedia/data services.

**Initial idea:** one global framework where **different radios** (e.g., **UTRA FDD/TDD**, **cdma2000**, **EDGE/DECT**) plug into a common **IMT-2000 core** via standardized interworking.

**What was the focus of early UMTS on? How is this reflected in the spectrum?**

**Early UMTS = Release 99 with UTRA FDD (W-CDMA) and the UTRAN architecture (NodeB/RNC)**, i.e., wideband CDMA radio + GSM-evolved core.

**Spectrum reflection:** the slide highlights UTRA FDD “up/down” at ~2.1 GHz (and TDD options)—large paired blocks for high-rate, wideband carriers; planning must handle **interference/cell breathing** in W-CDMA.

**What is the problem of the condition “coverage of the population in %”?**

It **overstates real availability**: you can hit high % by covering dense cities while **leaving rural/geographic areas, roads, and indoors** weak; and in W-CDMA, **cell breathing** can shrink coverage under load, so “% population” says little about **consistent service**.

**Why did most operators already using GSM pick UMTS as 3rd generation and cdmaOne operators cdma2000?**

**Migration continuity:** UMTS (UTRA) was the **3GPP** evolution of GSM (reusing spectrum strategy, core interworking, roaming ties); cdma2000 was the **3GPP2** evolution of cdmaOne (ANSI-41 core, same vendor/IPR path).

**Compare the handovers in GSM with the ones in UMTS. Why are they called hard or soft, respectively?**

**GSM = hard handover (break-before-make).** The MS leaves the old channel *before* traffic continues on the new one. Your slide shows the explicit “**Hard handover**” label and the classic steps: MS measurements → HO decision (BSC/MSC) → target resource allocation → **HO command** → **HO access** → **HO complete** → clear old. It also shows a configured **HO\_MARGIN** to avoid ping-pong.

**UMTS (W-CDMA) = soft/softer handover (make-before-break).** Thanks to code-division on the same carrier, the UE can keep **multiple radio links at once** (Active Set) and combine them; dropping the old link only after the new one is up. “**Softer**” HO is between sectors of the same NodeB; “**soft**” HO is between different NodeBs via the RNC (macrodiversity). (Conceptual—this mechanism is why CDMA systems can smooth fading and reduce drops.)

**How much is the core network aware of the type of handover?**

**GSM:** the BSC handles intra-BSC HOs; the MSC becomes involved for inter-BSC and inter-MSC HOs (your “4 types of handover” figure shows the BSC/MSC split). So awareness depends on scope.

**UMTS:** the UTRAN/RNC **encapsulates radio specifics** (cell-level mobility, combining, Active Set changes). The **core (CN)** mainly sees mobility when it crosses RNC (Iu) boundaries or systems; it’s **not involved** in soft/softer HOs internal to an RNC.

**What are the consequences of introducing IP-based packet transmission?**

**New packet core & protocols:** GSM→GPRS adds SGSN/GGSN and GTP/LLC/SNDCP; investment in new hardware, tunneling, and best-effort scheduling (PDCHs used only when packets are ready).

**Packet behavior:** variable delay/jitter, ARQ & scheduling overhead, header/tunnel overhead; bearer setup becomes **PDP context** instead of a circuit. (Your GPRS architecture/protocol slides illustrate this stack.)

**Toward all-IP:** in LTE this evolves into the **EPC** (MME/S-GW/P-GW, policy/QoS), fully IP bearer control and QoS policing.

**What is the key to higher data rates in UMTS?**

The slide highlights **HSDPA/HSUPA** → **HSPA+**: higher-order **modulation (16-QAM/64-QAM)**, **MIMO (2×2)**, and **dual/multi-carrier (DC/MC-HSPA)** aggregation. That’s the throughput jump.

**While the data rates increased over time, the delay remained more or less constant at about 100-150 ms from a mobile device to the base station. What problem arises if e.g. downloading content using TCP via these fast connections? Think of congestion avoidance/RTT!**

Consequences for TCP:

- **Slow-start takes longer at high BDP.** Throughput ramps **per RTT** (cwnd doubles each RTT); with a large bandwidth-delay product, it takes many RTTs to “fill the pipe,” so short downloads under-utilize the link.
- **Loss hurts more.** With long RTTs, timeouts and cwnd halving cost more wall-clock time; recovery is sluggish.

- **Ack-clocking & queueing.** Scheduler/ARQ jitter can bunch ACKs, causing burstiness and buffer growth; the sender may misinterpret delay as congestion.
- **Bottom line:** high peak **rate**  $\neq$  high **goodput** unless TCP is tuned (larger initial window, SACK, appropriate RCVBUF/SNDBUF), and radio scheduling/ARQ are gentle on latency.

### Why is LTE such a world-wide success?

**Radio flexibility & performance:** works from **700–2700 MHz** with **1.4–20 MHz** channels; **FDD/TDD, OFDMA (DL) / SC-FDMA (UL)**, **QPSK/16-QAM/64-QAM, MIMO**, mobility up to **350 km/h**—so operators could deploy in many bands with high spectral efficiency.

**Flat, all-IP architecture:** **eNodeB** folds RNC functions into the base station; **EPC** (MME/S-GW/P-GW/HSS/PCRF) replaces the GPRS core and is optimized for IMS/VoLTE—lower latency, simpler scaling.

**Interference/capacity features:** **soft frequency reuse** (power partitioning between cell center/edge) and **X2** links for fast eNB-to-eNB coordination.

### Name high-level differences between LTE and GSM/GPRS/UMTS! What do they have in common

**Different:** LTE uses **OFDMA/SC-FDMA** (not TDMA or W-CDMA), **flat IP core (EPC)** (no circuit switching), **eNodeB (RAN control at the edge)**, tight **QoS scheduling** in the eNodeB, and **X2/S1** for handover.

**Common:** SIM-based auth (USIM on UICC), PLMN roaming, bearer/QoS concepts and subscriber databases (**HSS ~ HLR** evolution), interworking to external PDNs.

### What is the idea of soft frequency reuse?

**Center:** use **all sub-bands at lower power**; **edge:** reserve certain sub-bands at **higher power** → raises edge SINR while keeping good center capacity (a form of interference coordination).

### Compare the architecture of LTE/UMTS/GSM – what is new, what is basically the same?

**New:** **eNodeB** = NodeB+RNC; **EPC** nodes and **policy control (PCRF)**; **X2** mesh between eNodeBs; **S1** to the core; **IP-only bearers**.

**Same in spirit:** subscriber DB (**HSS**), mobility anchoring (**S-GW** analogous to GPRS mobility anchor), gateways to the Internet (**P-GW ~ GGSN** role).

### What is the purpose of the X2 connections between eNodeBs? How can LTE support seamless / lossless handover?

**Purpose:** direct **eNB↔eNB** signaling (**X2-C**) and user-plane forwarding (**X2-U**) for coordination, load, and **fast handover** without going up to the core when possible.

**Seamless/lossless HO:** via **X2 HO** (buffer/forward PDCP packets between eNBs); fall back to **S1 HO** through MME/S-GW if needed. (Slides show both **X2** and **S1** HO paths.)

### How can LTE guarantee data rates, delays?

**Bearer-level QoS:** **GBR/Non-GBR bearers** with **QoS authorization** in **PCRF**; **eNodeB scheduling** enforces per-bearer QoS, admission control, and prioritization. **Flat RAN + short TTI:** RAN decisions at the eNodeB keep latency low (slides note lower RTT vs 3G).

### Compare the multiplexing/multiple access in LTE with UMTS/GSM – differences/similarities?

**LTE:** **OFDMA (DL)**, **SC-FDMA (UL)** over **1.4–20 MHz** channels; flexible TDD/FDD. , **UMTS:** wideband **CDMA** (soft/softer HO, cell breathing), **GSM/GPRS:** **FDMA+TDMA** (8 slots @ 200 kHz).

**Evolutionary in LTE:** retains 3GPP's bearer/QoS and mobility ideas, but makes the **RAN flatter and fully packet-switched**, with **wider/aggregated carriers** later in LTE-A/Pro.

## How does LTE reflect the variety of different demands?

**Spectrum agility** (many bands, 1.4–20 MHz), **MIMO**, **cell sizes** from <1 km to ~100 km, **high mobility**, and dedicated IoT profiles (NB-IoT/LTE-M, shown as “LTE for IoT”).

### 4.6 Review exercises

- 1 Name some key features of the GSM, DECT, TETRA, and UMTS systems. Which features do the systems have in common? Why have the three older different systems been specified? In what scenarios could one system replace another? What are the specific advantages of each system?
- 2 What are the main problems when transmitting data using wireless systems that were made for voice transmission? What are the possible steps to mitigate the problems and to raise efficiency? How can this be supported by billing?
- 3 Which types of different services does GSM offer? Give some examples and reasons why these services have been separated.
- 4 Compared to the TCHs offered, standard GSM could provide a much higher data rate (33.8 kbit/s) when looking at the air interface. What lowers the data rates available to a user?
- 5 Name the main elements of the GSM system architecture and describe their functions. What are the advantages of specifying not only the radio interface but also all internal interfaces of the GSM system?
- 6 Describe the functions of the MS and SIM. Why does GSM separate the MS and SIM? How and where is user-related data represented/stored in the GSM system? How is user data protected from unauthorized access, especially over the air interface? How could the position of an MS (not only the current BTS) be localized? Think of the MS reports regarding signal quality.
- 7 Looking at the HLR/VLR database approach used in GSM – how does this architecture limit the scalability in terms of users, especially moving users?
- 8 Why is a new infrastructure needed for GPRS, but not for HSCSD? Which components are new and what is their purpose?
- 9 What are the limitations of a GSM cell in terms of diameter and capacity (voice, data) for the traditional GSM, HSCSD, GPRS? How can the capacity be increased?
- 10 What multiplexing schemes are used in GSM and for what purpose? Think of other layers apart from the physical layer.
- 11 How is synchronization achieved in GSM? Who is responsible for synchronization and why is it so important?
- 12 What are the reasons for the delays in a GSM system for packet data traffic? Distinguish between circuit-switched and packet-oriented transmission.
- 13 Where and when can collisions occur while accessing the GSM system? Compare possible collisions caused by data transmission in standard GSM, HSCSD, and GPRS.
- 14 Why and when are different signaling channels needed? What are the differences?
- 15 How is localization, location update, roaming, etc. done in GSM and reflected in the data bases? What are typical roaming scenarios?
- 16 Why are so many different identifiers/addresses (e.g., MSISDN, TMSI, IMSI) needed in GSM? Give reasons and distinguish between user-related and system-related identifiers.
- 17 Give reasons for a handover in GSM and the problems associated with it. What are the typical steps for handover, what types of handover can occur? Which resources need to be allocated during handover for data transmission using HSCSD or GPRS respectively? What about QoS guarantees?
- 18 What are the functions of authentication and encryption in GSM? How is system security maintained?
- 19 How can higher data rates be achieved in standard GSM, how is this possible with the additional schemes HSCSD, GPRS, EDGE? What are the main differences of the approaches, also in terms of complexity? What problems remain even if the data rate is increased?
- 20 What limits the data rates that can be achieved with GPRS and HSCSD using real devices (compared to the theoretical limit in a GSM system)?
- 21 Using the best delay class in GPRS and a data rate of 115.2 kbit/s – how many bytes are in transit before a first acknowledgement from the receiver could reach the sender (neglect further delays in the fixed network and receiver system)? Now think of typical web transfer with 10 kbyte average transmission size – how would a standard TCP behave on top of GPRS (see chapters 9 and 10)? Think of congestion avoidance and its relation to the round-trip time. What changes are needed?
- 22 How much of the original GSM network does GPRS need? Which elements of the network perform the data transfer?
- 23 What are typical data rates in DECT? How are they achieved considering the TDMA frames? What multiplexing schemes are applied in DECT and for what purposes? Compare the complexity of DECT with that of GSM.
- 24 Who would be the typical users of a trunked radio system? What makes trunked radio systems particularly attractive for these user groups? What are the main differences to existing systems for that purpose? Why are trunked radio systems cheaper compared to, e.g., GSM systems for their main purposes?
- 25 Summarize the main features of third generation mobile phone systems. How do they achieve higher capacities and higher data rates? How does UMTS implement asymmetrical communication and different data rates?
- 26 Compare the current situation of mobile phone networks in Europe, Japan, China, and North America. What are the main differences, what are efforts to find a common system or at least interoperable systems?
- 27 What disadvantage does OVSF have with respect to flexible data rates? How does UMTS offer different data rates (distinguish between FDD and TDD mode)?
- 28 How are different DPDCHs from different UEs within one cell distinguished in UTRA FDD?
- 29 Which components can perform combining/splitting at what handover situation? What is the role of the interface I<sub>ur</sub>? Why can CDMA systems offer soft handover?
- 30 How does UTRA-FDD counteract the near-far effect? Why is this not a problem in GSM?



#### 4. Telecommunication systems

- 4.1 Key features: GSM (wide area coverage, bandwidth 9.6-50 kbit/s, voice, SMS, MMS), DECT (local coverage, voice, data, high density), TETRA (regional coverage, ad-hoc mode, very fast connection set-up, group call, voice, data, very robust), UMTS (medium coverage, higher data rates 384 kbit/s, flexible bandwidth assignment). Common features are traditional voice support (circuit switched), integration into classical fixed telecommunication network, ISDN core network. The systems have different, unique properties: GSM has wide area coverage, TETRA ad-hoc mode and fast connection setup, DECT can support high user densities. If allowed from licensing GSM could replace DECT, if modified GSM can replace TETRA (e.g., GSM-Rail) – under certain conditions (GSM does not offer an ad-hoc mode). But also UMTS has specific advantages – higher data rates compared to classical GSM (but lower coverage) and higher coverage than WLANs (but lower data rates).
- 4.2 Systems optimised for voice transmission support certain fixed data rates and operate circuit switched. Data transmission happens quite often spontaneous with varying data rates. Thus either too much bandwidth is reserved to accommodate the maximum expected data rate or data transmission experiences long delays due to connection setup. One possible step towards the support of data transmission is the introduction of packet switched services as known from the Internet. An example is GPRS in GSM. Instead of time-based billing providers can now bill based on volume (however, application based billing would make even more sense as customers are not interested in bytes but useful applications).
- 4.3 The three big categories are bearer, tele, and supplemental services. Separation of services supports phased introduction of services and separation of concerns: network providers, service providers, device manufacturers etc. can focus on certain sets of services (e.g., tele services between terminals) and rely on certain interfaces to other services (e.g., to the underlying bearer services).
- active (registered). If many users move between location areas updates have to take place, i.e., the HLR always gets the information about the new VLR. These updates happen independently on the users' activity (data transmission, calls etc.). For standard scenarios – most users stay most of the time within their location area – the 2-level hierarchy works well. However, if, e.g., many tourists move frequently the updating process puts some load on the network as the HLR in the home network of the tourists always requires update information – probably around the globe. More levels of hierarchy could improve scalability but also raises complexity.
- 1.8 HSCSD still operates circuit switched as CSD does. It "simply" combines several connections. GPRS introduces a new paradigm in GSM, packet switching. Basically, the core network needs routers handling the packet stream. These routers (SGSN, GGSN) operate on IP and rely on the traditional GSM network for user localisation. Another new component located at the HLR is a registry for subscribed GPRS services. Furthermore, the system has to set up a context for each active user, account transmitted data, assign IP addresses etc.
- 1.9 Traditional GSM has cell diameters of up to 70 km, i.e., a user may have a maximum distance of 35 km to the base station. This limitation is not because of too strong attenuation, but because of the delay the signals experience. All signals must arrive synchronised at the base station, timing advance adjust the sending point (the further away a terminal is the earlier it has to send its data). With some tricks the diameter can be doubled. The capacity is limited by the number of channels \* number of time slots – signalling overhead. The number of channels is operator and regulation dependent. The capacity is independent of the usage of GSM/CSD, HSCSD or GPRS – all three systems use the same basic frame structure and modulation. New modulation schemes can offer higher capacity, EDGE is an example. Furthermore, systems like GPRS offer different levels of error protection – this may increase user data rates under good propagation conditions, but does not increase the system capacity.
- 4.4 Main reason is the forward error correction to mitigate transmission errors. Furthermore, bandwidth is needed for signalling, guard spaces.
- 4.5 See figure 4.4. Specifying all (or at least many) internal interfaces allows for a larger variety of vendors. As long as vendors stay with the standardised interfaces equipment of different vendors can be combined and network operators are not completely dependent from one manufacturer. However, reality often looks different and network operators often use only equipment from one or two vendor(s).
- 4.6 The MS contains all device related functions: device ID, coders/decoders, radio etc. The SIM contains subscriber related functions and data: authentication, PIN, user id etc. This separation helps changing phones while keeping personal data: users simply insert their SIM in a new mobile phone and can use, e.g., their personal phone book, PIN etc. Exceptions are so-called SIM locked phones – in this case a mobile phone accepts only a certain SIM. However, this is rather a marketing than technical reason. Besides the SIM also the mobile phone itself can store user-related data. Additional user-related data is stored in the VLR responsible for the location area a user is currently in and the HLR of the network operator the user has a contract with. User data is protected in several ways: authentication centres are protected parts of the HLR residing at the network operator. Inside the core network only temporary identifiers are used, data is encrypted over the air interface (weak, but still encrypted), and the content of the SIM is protected via a PIN (some cards destroy themselves after being attacked too many times). Localisation could be terminal assisted: the terminal could gather the current signal strength from all surrounding base stations. Furthermore, using the time of arrival helps calculating the distance. Reflection and attenuation makes the calculation more difficult.
- 4.7 GSM uses only two levels of hierarchy: Network operators store all user related information in the HLR and all information related to visitors within a certain location area in a VLR. Capacities of HLRs is up to some million customers, that of VLRs up to a million. I.e., within the location area a maximum of, e.g., one million users can be
- 4.10 GSM uses SDM, FDM and TDM:
- SDM: Operators design the cell layout, place base stations and reuse frequencies according to certain cluster patterns.
  - FDM: Regulation authorities assign channels to operators, operators assign channels to base stations, and base stations assign a certain channel to a terminal during data transmission.
  - TDM: Base stations assign a time-slot or several time-slots to a terminal for transmission.
- 4.11 The BSS has to create a frame structure. Terminals listen into the medium, receive signals over broadcast channels and synchronise to the frame structure. Within each time-slot during transmission a midamble further improves synchronisation. The terminal itself is responsible for precise synchronisation within the cell. This is very important in TDM systems as otherwise neighbouring data may be destroyed.
- 4.12 Examples for delays in packet transmission:
- CS: connection setup (some seconds), FEC coding/decoding and interleaving (about 100 ms), propagation delay (some ms)
  - PS: channel access (depending on the current load), FEC coding/decoding and interleaving (about 100 ms), propagation delay and routing (some ms). Experiments show that packets in GPRS may experience heavy delays due to channel access delays: 200-500 ms for 128 byte packets, several seconds for 1-4 kbyte packets.
- 4.13 Besides problems due to interference, collisions in GSM systems can only occur during connection setup. Terminals have to access the base station using a slotted Aloha scheme for the layer 2 signalling connection. During this connection attempt several terminals may collide and have to repeat the connection attempt. During data transmission or voice call no collision can occur. Data transmission in standard GSM (CSD) behaves just as voice calls. HSCSD has the additional problem of requesting

several channels. These may be occupied. However, this does not cause a collision but a simple denial of the connection request for several channels. Channel assignment and release is handled dynamically in GSM systems. For GPRS, too, data transmission can not cause a collision as the terminal wanting to transmit has to request time-slots first. After the assignment of time-slots the terminal may access these slots without further collisions. Depending on the current load, not too many slots may be available; however, network operators try to offer at least one slot per cell for GPRS traffic to offer a minimum data rate.

- 4.14 GSM comprises many different channels for signalling control data. If no traffic channel (TCH) exists, an MS uses an SDCCH for signalling, e.g., authentication and registration data required prior to TCH establishment. TCH and SDCCH use an SACCH for signalling channel quality/signal strength. If a TCH exists and more signalling is required (e.g., during handover), an MS uses a FACCH, which is located in the time-slots otherwise used by the TCH.

- 4.15 The GSM system only stores the current location area for a user in the VLR. Each time a user changes the location area this change is reflected in the VLR. Additionally, periodic updates are possible. Roaming includes changing the network operator. This can happen within the same country (national roaming) or when going to another country (international roaming). The latter is the most common scenario as national roaming typically involves direct competitors. Prerequisite are roaming agreements between the different operators. The HLR always stores the current VLR for the user, no matter if inside the own or inside a foreign network. Precise localisation of users is performed during call setup only (paging within the location area).

- 4.16 Users of the GSM systems work with telephone numbers. That is all users should see. These phone numbers are completely independent of the current location of the user. The system itself needs some additional information; however, it must not reveal the identity of users. The international identification of users is done with the

- *LMSI*: An additional local mobile subscriber identity (32 bit) can be used by the VLR/HLR for fast subscriber look-up.
- *CI*: Within a LA each cell has a unique cell identifier (16 bit). Thus, the tuple (LAI, CI) uniquely identifies a cell worldwide (global cell identity).
- *BSIC*: The base transceiver station identity code identifies base stations (6 bit) and consists of a 3 bit network colour code and a 3 bit base transceiver station colour code.
- All MSCs, VLRs and HLRs have unique ISDN numbers for identification.

- 4.17 The typical reason for a handover is a weaker signal from the current base station compared with a neighbouring base station. Another reason could be the current load situation: the network could decide to offload some users from a crowded cell. For the typical steps and types of handover see figures 4.11-4.13. For HSCSD to succeed the same resources are needed in the new cell as were available in the old one. I.e., there must be enough time-slots available to handle the same number of simultaneous connections. Otherwise the available bandwidth will decrease. Sure the probability of having several channels available is much lower than having a single channel. For GPRS data rates fluctuate anyway depending on the current load. The same happens during and after handover. Without pre-reservation neither HSCSD nor GPRS can give any QoS guarantees. There is not even a QoS guarantee for a voice call – if the next cell is already completely booked the connection will break upon entering this cell.

- 4.18 The first step is the authentication of the user against the SIM. This is done using a simple PIN. Then, the SIM authenticates itself against the GSM system. This second authentication is much stronger compared to the PIN. This is because the operator is not really interested in who is using the system as long as it is a valid and paying customer. Authentication with the system uses a challenge response scheme with a shared secret on the SIM and in the AuC. Neither the SIM nor the AuC will transmit

IMSI (=country code + network code + subscriber ID). During operation within a location area, only a temporary identifier, the TMSI is needed. This hides the identity of a user. The TMSI is not forwarded to the HLR. These are already some examples for identifiers; however, GSM provides some more:

- *IMEI*: MS identification (like a serial number); consists of type approval code (centrally assigned), final assembly code, serial number, and spare (all three manufacturer assigned).
- *IMSI*: Subscriber identification, stored in the SIM. Consists of the mobile country code (3 digits, e.g., 262 for Germany), the mobile network code (2 digits, e.g., 01 for the German T-Mobile), and the mobile subscriber identification number (10 digits). The mobile network code together with the mobile subscriber identification number forms the national mobile subscriber identity.
- *MSISDN*: Mobile subscriber ISDN Number, i.e., the phone number, assigned to a subscriber, not a telephone! The MSISDN is public, not the IMSI nor the mapping MSISDN-IMSI. An MSISDN consists of the country code (up to 3 digits, e.g., 49 for Germany), the national destination code (typically 2 or 3 digits), and the subscriber number (up to 10 digits).
- *MSRN*: The mobile station roaming number is a temporarily assigned, location based ISDN number. The VLR assigns MSRNs and forwards them to the HLR/GMSC for call forwarding. Assignment happens either upon entering a new LA or upon request from the HLR (call-setup).
- *LAI*: The location area identity describes the LA of a network operator. It consists of a country code (3 digits), a mobile network code (2 digits), and a location area code (16 bit). The LAI is broadcasted on the BCCH for LA identification.
- *TMSI*: The VLR currently responsible for an MS can assign a 32 bit temporary mobile subscriber identity to an MS with a SIM. The tuple (TMSI, LAI) uniquely identifies a subscriber. Thus, for ongoing communication IMSI is replaced by (TMSI, LAI).

this secret over the air or reveal it to customers. Encryption only takes places between the MS and the BSS. GSM does not provide strong encryption end-to-end or MS to the gateway into the fixed network. System designers decided for over-the-air encryption only as they thought that the system itself is trustworthy. Thus, authentication of base stations against MSs was neglected, too. This opened ways to fake base stations. UMTS introduces full authentication of all components.

- 4.19 The classical data rate of GSM is 9.6 kbit/s. Using less FEC 14.4 kbit/s are available, too. These data rates are achievable using a single time-slot per frame in a certain channel. HSCSD combines several time-slots but leaves coding untouched. GPRS can dynamically use several time-slots per frame plus offers 4 different coding schemes that allow for higher data rates per slot. EDGE finally introduces another modulation scheme (PSK) in addition to GMSK, which offers even higher data rates under good propagation conditions. Only EDGE can really increase the capacity of a GSM cell. Independent of the coding and modulation schemes the complexity of handover signalling, handover delay and high delay due to coding/interleaving remain.

- 4.20 Real devices can (currently) not offer all data rates specified in the standards. While the standards in principle specify devices that use all 8 time-slots in both directions, real devices can often not send and receive at the same time. Furthermore, older devices even need some time to switch from sending into receiving mode, thus wasting another slot or even several slots. Additionally, current GPRS phones often do not offer all coding schemes.

- 4.21 The delay is specified between the MS and the exit point of the GPRS core network. The best average delay is 0.5 s. Assuming a data rate of 115.2 kbit/s (a common rate using serial adapters connected to the mobile phone) and a delay of 2\*0.5 sec, 115.2 kbit = 14.4 kbyte are in transit. TCP was made for streaming larger amounts of data, i.e., file transfers etc. TCP allows for fair sharing of bandwidth as soon as it is in stable state. This requires the reception of acknowledgements, the adaptation of



- sending windows and thresholds. However, if the whole transfer is 10 kbyte only, TCP either never gets an acknowledgement back during transmission to adapt sender characteristics (only if the initial sending window is large enough), or TCP wastes bandwidth by using a too small starting sending window (standard case). Real measurements with GPRS exhibit high latencies (examples are round trip times for different packet sizes, class 8 mobile phone): 0.8 s/64 byte, 1.4 s/128 byte, 2.2 s/1024 byte, 2.9 s/2048 byte, and 4.8 s/4096 byte. Additionally, measurements show high jitter. Under these conditions, TCP performs poorly. Chapter 9 lists several proposed changes to TCP (e.g., large initial sending window).
- 4.22 GPRS still needs the classical CS core for localisation, authentication etc. However, for data transfer the MSCs are not needed any more. The routers in the PS core (SGSN and GGSN) perform data forwarding (see figure 4.16).
- 4.23 DECT offers 120 full duplex channels, each with a standard rate of 32 kbit/s (unprotected). DECT applies TDM for structuring the frames and multiplexing users (24 slots per frame, typ. 12 up/12 down link). Furthermore, FDM is applied to increase capacity (several DECT cells at the same location, 10 channels). Users can also apply SDM by placing access points further apart. All the multiplexing schemes together result in very high capacities of the system, which is needed, e.g., in office buildings. Compared to GSM the system is simpler. Although data bases have been defined, too, typical DECT systems consist of a simple base station and several mobile devices. Most scenarios do not require complicated handover (although possible in DECT). Most systems furthermore do not need accounting and billing mechanisms as they are simply connected to the fixed phone network or a PBX.
- 4.24 Police, fire brigades, ambulances, disaster relief teams, public transportation authorities, taxi drivers, etc. are typical users of trunked radio systems. Trunked radio systems are attractive because of special features like very fast connection setup (sub second), group calls, paging, high robustness, cheap operation, reliable and fast messaging, and ad-hoc capabilities. Existing systems for these special purposes are
- Japan: Two different 3G systems are available in Japan. NTT DoCoMo uses a variation of UMTS in their W-CDMA system marketed as FOMA. KDDI deploys a cdma2000 system, which is 3G from the version 1xEV-DO on.
  - China: While most 2G users today use GSM creating the biggest national market for this system, it may be speculated that UMTS will be a major 3G system in China, too, as this system can easily reuse the existing core network in its Release 99. The Chinese development TD-SCDMA was incorporated into UMTS (UTRA/TDD, slow chipping option, Release 4). However, it is currently not clear when and if this system will be deployed. There are also some cdma-operators in China which might opt for cdma2000.
  - North America: The situation in the US and Canada is quite unclear. Already today many systems exist in parallel without a clear winner (compared to GSM in Europe). Furthermore, licensing of 3G spectrum takes a long time and the availability of spectrum is not clear yet. Thus, it could be the case that EDGE enhanced systems (TDMA and GSM) will be deployed offering higher data rates with EGPRS compared to today's networks. The cdma-operators will go for cdma2000.
- 1.27 QVSF offers only certain fixed data rates (certain multiples of 15 kbit/s). If users want to send with a data rate in-between the system either drops data (which can be recovered using FEC) or inserts dummy data. In the FDD mode adjusting the spreading factor is the only way for offering different data rates. TDD offers additionally the possibility of requesting more or less slots for up or downlink.
- 1.28 The spreading codes can always be the same in UTRA FDD to lower the system complexity. However, each UE has an individual scrambling code that is quasi-orthogonal to other scrambling codes. In UTRA TDD, scrambling is cell specific.
- 1.29 The important characteristic of combining/splitting is that it is never performed inside the (traditional) core network. I.e., the MSCs do not notice anything from the new possibilities offered by CDMA (reception of data via more than one base station).
- still often analogue systems operating on special frequencies without strong encryption. This makes it very difficult to cooperate for, e.g., fire brigades, the police and ambulances during disaster relief operations – the teams have to exchange equipment in order to be able to communicate. Trunked radio systems can be cheaper compared to GSM as they can have higher coverage with fewer base stations due to the lower expected load. Furthermore, complex billing and accounting mechanisms are quite often not needed.
- 4.25 Main features: higher and more flexible data rates, better voice quality due to new codecs, usage of CDMA (in almost all systems), operation at 2 GHz. Higher cell capacities and higher data rates are mainly achieved by more powerful modulation schemes, better codecs with higher compression rates for voice, CDMA as additional multiplexing scheme, and more powerful devices (more precise power adaptation, utilisation of multipath propagation, ...). UMTS implements asymmetrical data rates and different data rates in the same direction via different spreading factors. As the chipping rate of UMTS is always constant, data rates depend on the spreading factor. The more the data is spread the lower the data rate is.
- 4.26 Currently, the situation is not absolutely clear as the different countries are in different stages implementing 3G systems. Right now no one believes in a common worldwide system, not even the same frequencies are available everywhere:
- Europe: After a much discussed licensing process (beauty contests and auctions) many operators are currently deploying 3G systems. Some operators already dropped out, some filed bankruptcy. All operators for 3G will use UMTS, in the beginning the UTRA/FDD mode only (no one knows when and if UTRA/TDD will be deployed). Although licensing did not prescribe the usage of UMTS, there were only a few operators thinking of different systems in the beginning. Start of the system was 2002, 50% of the population should have access to UMTS in 2005 (in Germany).
- Depending on the location of the handover (between two antennas at the same Node B, between two Node Bs, between two RNCs) the Node Bs or the serving RNC have to perform splitting/combination. The interface  $I_{ur}$  is needed to transfer data between RNCs for combination/splitting without any interaction with the CN. For CDMA receiving signals from different base stations looks like multipath propagation. The rake receivers can thus handle both. The handover is then as soft as a change in the strongest signals in a multipath scenario. TDMA/FDMA systems like GSM cannot do this because the currently used time-slot and/or frequency may not be available in the next cell.
- 4.30 The terminals have to measure and adapt their transmission power 1500 times per second in UTRA/FDD to achieve equal signal strength at the base station. In GSM this is no problem as it never happens that two stations send at the same time on the same frequency.

### Example 2.1

If a total of 33 MHz of bandwidth is allocated to a particular FDD cellular telephone system which uses two 25 kHz simplex channels to provide full duplex

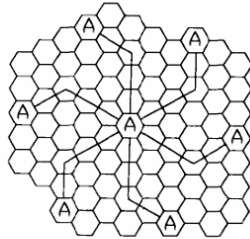


Figure 2.2  
Method of locating co-channel cells in a cellular system. In this example,  $N = 19$  (i.e.,  $i = 3, j = 2$ ).  
[Adapted from [Oet83] © IEEE].

voice and control channels, compute the number of channels available per cell if a system uses (a) 4-cell reuse, (b) 7-cell reuse (c) 12-cell reuse. If 1 MHz of the allocated spectrum is dedicated to control channels, determine an equitable distribution of control channels and voice channels in each cell for each of the three systems.

### Solution to Example 2.1

Given:  
Total bandwidth = 33 MHz  
Channel bandwidth = 25 kHz  $\times$  2 simplex channels = 50 kHz/duplex channel  
Total available channels =  $33,000/50 = 660$  channels  
(a) For  $N = 4$ ,  
total number of channels available per cell =  $660/4 \approx 165$  channels.  
(b) For  $N = 7$ ,  
total number of channels available per cell =  $660/7 \approx 95$  channels.  
(c) For  $N = 12$ ,  
total number of channels available per cell =  $660/12 \approx 55$  channels.

A 1 MHz spectrum for control channels implies that there are  $1000/50 = 20$  control channels out of the 660 channels available. To evenly distribute the control and voice channels, simply allocate the same number of channels in each cell wherever possible. Here, the 660 channels must be evenly distributed to each cell within the cluster. In practice, only the 640 voice channels would be allocated, since the control channels are allocated separately as 1 per cell.  
(a) For  $N = 4$ , we can have 5 control channels and 160 voice channels per cell. In practice, however, each cell only needs a single control channel (the control

channels have a greater reuse distance than the voice channels). Thus, one control channel and 160 voice channels would be assigned to each cell.  
(b) For  $N = 7$ , 4 cells with 3 control channels and 92 voice channels, 2 cells with 3 control channels and 90 voice channels, and 1 cell with 2 control channels and 92 voice channels could be allocated. In practice, however, each cell would have one control channel, four cells would have 91 voice channels, and three cells would have 92 voice channels.  
(c) For  $N = 12$ , we can have 8 cells with 2 control channels and 53 voice channels, and 4 cells with 1 control channel and 54 voice channels each. In an actual system, each cell would have 1 control channel, 8 cells would have 53 voice channels, and 4 cells would have 54 voice channels.

### Example 2.2

If a signal to interference ratio of 15 dB is required for satisfactory forward channel performance of a cellular system, what is the frequency reuse factor and cluster size that should be used for maximum capacity if the path loss exponent is (a)  $n = 4$ , (b)  $n = 3$ ? Assume that there are 6 co-channels cells in the first tier, and all of them are at the same distance from the mobile. Use suitable approximations.

### Solution to Example 2.2

(a)  $n = 4$   
First, let us consider a 7-cell reuse pattern.  
Using equation (2.4), the co-channel reuse ratio  $D/R = 4.583$ .  
Using equation (2.9), the signal-to-noise interference ratio is given by  
 $S/I = (1/6) \times (4.583)^4 = 75.3 = 18.56$  dB.  
Since this is greater than the minimum required  $S/I$ ,  $N = 7$  can be used.  
  
(b)  $n = 3$   
First, let us consider a 7-cell reuse pattern.  
Using equation (2.9), the signal-to-interference ratio is given by  
 $S/I = (1/6) \times (4.583)^3 = 16.04 = 12.05$  dB.  
Since this is less than the minimum required  $S/I$ , we need to use a larger  $N$ .  
Using equation (2.3), the next possible value of  $N$  is 12, ( $i = j = 2$ ).  
The corresponding co-channel ratio is given by equation (2.4) as  
 $D/R = 6.0$ .

## 2.9 Problems

1. Prove that for a hexagonal geometry, the co-channel reuse ratio is given by  $Q = \sqrt{3N}$ , where  $N = i^2 + ij + j^2$ . (Hint: use the cosine law and the hexagonal cell geometry).
2. Show that the frequency reuse factor for a cellular system is given by  $k/S$ , where  $k$  is the average number of channels per cell and  $S$  is the total number of channels available to the cellular service provider.
3. A cellular service provider decides to use a digital TDMA scheme which can tolerate a signal-to-interference ratio of 15 dB in the worst case. Find the optimal value of  $N$  for (a) omni-directional antennas, (b)  $120^\circ$  sectoring, and (c)  $60^\circ$  sectoring. Should sectoring be used? If so, which case ( $60^\circ$  or  $120^\circ$ ) should be used? (Assume a path loss exponent of  $n = 4$  and consider trunking efficiency).
4. If an intensive propagation measurement campaign showed that the mobile radio channel provided a propagation path loss exponent of  $n = 3$  instead of 4, how would your design decisions in Problem 2.3 change? What is the optimal value of  $N$  for the case of  $n = 3$ ?
5. For a  $N = 7$  system with a  $Pr[\text{Blocking}] = 1\%$  and average call length of 2 minutes, find the traffic capacity loss due to trunking for 57 channels when going from omni-directional antennas to  $60^\circ$  sectorized antennas. (Assume that blocked calls are cleared and the average per user call rate is  $\lambda = 1$  per hour.

Using equation (2.3) the signal-to-interference ratio is given by  
 $S/I = (1/6) \times (6)^3 = 36 = 15.56$  dB.  
Since this is greater than the minimum required  $S/I$ ,  $N = 12$  can be used.